From reviews to decisions.

What should an Amazon vendor improve next—and why?

01 Find the complaint

02 Verify the evidence

03 Recommend a test

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Final project | Amazon Reviews 2023

A star rating is a signal. Not a diagnosis.

Vendors need a specific, testable next step—not another generic sentiment summary.

PRODUCT IMPROVEMENT

What material, fit, component or use condition should the vendor investigate?

LISTING IMPROVEMENT

Which promise, sizing detail or limitation should the listing explain more clearly?

Deliverable → proposed change + review evidence + verification check

Our data: Clothing, Shoes & Jewelry

One Amazon dataset category, spanning three recognizable product families.



CLOTHING

Fit · fabric · sizing



SHOES

Comfort · support · durability



JEWELRY

Material · appearance · quality

DATASET SCOPE ≠ DEMO MIX

The five selected demonstration products are apparel and footwear.

A reproducible, intentionally limited sample

We sampled large Amazon files, joined reviews to metadata, and applied fixed eligibility rules.

8,535
Matched analytical reviews

5,221
Distinct parent products

640 MiB
Random byte-block sample

01 Clean + deduplicate

02 Join by parent ASIN

03 Cap at 200 / parent

04 Apply eligibility

Demo: 5 selected products • 246 reviews • 9 eligible parents

Sampling is reproducible, but nearby records cluster; results are not a representative census.

The hybrid pipeline: measure → interpret → verify

1

PYTHON

Attributes · clause-level VADER · low-star lift

2

RETRIEVE

TF-IDF matching to listing language · LDA themes

3

LLM

Interpret difficult reviews · draft proposed actions

4

VALIDATE

Check review IDs and verbatim source evidence

LOW-STAR LIFT P(attribute mentioned | 1–2 stars) / P(attribute mentioned)

We do not automatically trust model output.

The LLM only contributes when its evidence passes strict checks.

79
Selected for semantic review

24
Accepted annotations

55
Lexical fallback retained

30.4% **Evidence acceptance rate ≠ model accuracy**

Latest run: 30,675 tokens / ~$0.009 • Human-label accuracy evaluation remains unfinished.

Five products. Four action-worthy signals.

PRODUCT	REVIEWS	LOW STARS	MAIN SIGNAL	VENDOR NEXT STEP
SIMARI	20	25.0%	Durability · 6	Test sole / fabric durability
SOLLBEAM	149	12.8%	Support · 16	Test support; clarify limits
SHAPERMINT	23	39.1%	Listing mismatch · 3	Clarify fit and rolling claims
REEF	34	11.8%	Straps · 3	Test strap material / stitching
FRUIT OF THE LOOM	20	20.0%	No repeated signal	Collect more evidence first

16 proposals across 4 products • 10 GPT-generated + 6 evidence-derived fallbacks

AHA: the hybrid method found hidden complaints.

SHAPERMINT • negative distinct-review mentions

LISTING MISMATCH

1 Lexical → 3 Hybrid

FIT / SIZING

3 Lexical → 6 Hybrid

REVIEW EVIDENCE

“Advertised as high waisted and is not”

Listing mismatch lift: 2.56

BUSINESS ACTION → verify fit; clarify sizing and rolling behavior

More detections ≠ proven accuracy. 12 of 23 reviews had accepted semantic annotations.

What the vendor actually receives

A traceable decision-support output—not just a score.

01

ISSUE

Recurring complaint and supporting counts

02

EVIDENCE

Exact review text + related listing wording

03

ACTION

Proposed product test or listing clarification

04

CHECK

What the vendor must verify before acting

Sometimes the correct recommendation is: gather more evidence.

From hypothesis to measurable improvement

SELECT

Choose parent or child ASIN

REVIEW

Inspect cited complaint evidence

TEST

Verify and implement one change

MEASURE

Monitor complaints, returns, ratings

THE TAKEAWAY

Evidence-grounded hypotheses, not proven causal effects.

Limits: small clustered samples · possible historical listing mismatch · partial semantic coverage · no completed human validation.